

Al-Aqsa University
Faculty of Computers and IT
Engineering and Networks Dept.



جامعة الأقصى
كلية الحاسبات وتكنولوجيا المعلومات
قسم الهندسة والشبكات

Operating Systems for Intelligent Systems (ENGS2306)

Assignment 2

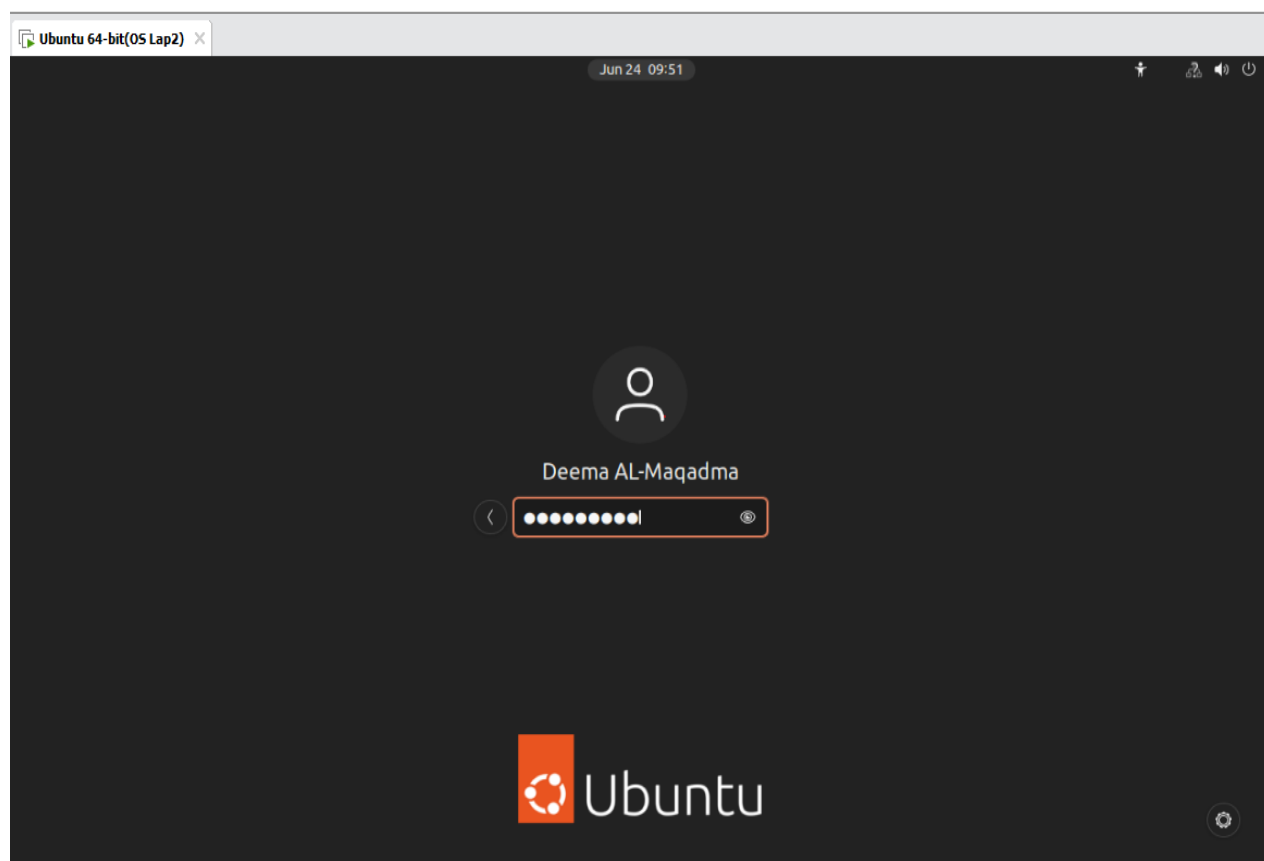
Linux File System and Permissions Assignment Solution

Presented to: **Eng. Mohammed AL-Hams**

Student name: **Deema Mohammed AL-Maqadma**

Student number: 2320230766

Semester: Second Semester 2024/2025



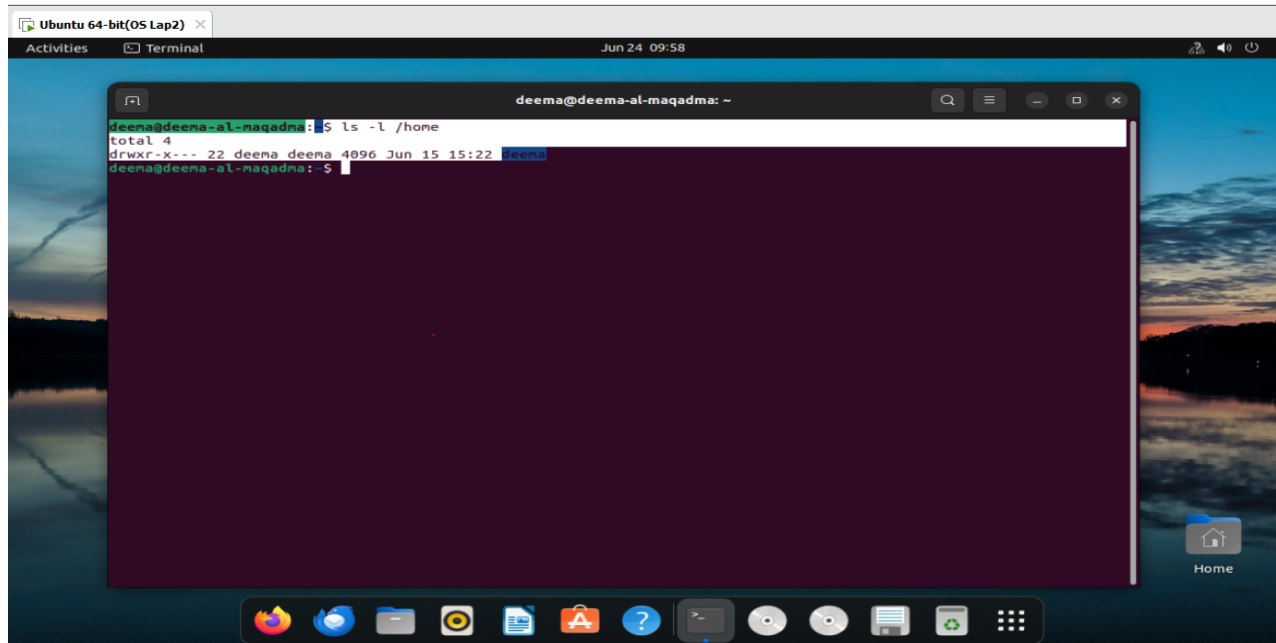
```
deema@deema-al-maqadma: ~  
deema@deema-al-maqadma:~$ echo "  
> ---> Deema Mohammed AL-Maqadma  
> Assignment ( 2 )  
> ---> Eng. Mohammed AL-Hams  
> "  
  
---> Deema Mohammed AL-Maqadma  
Assignment ( 2 )  
---> Eng. Mohammed AL-Hams  
  
deema@deema-al-maqadma:~$
```

Part 1: Exploring the File System /

1. File System Structure Exploration :

List contents of /home with details

ls -l /home : Function: Displays files and directories in /home with ownership and permission details, Expected output: Shows user directories with their permissions and owners

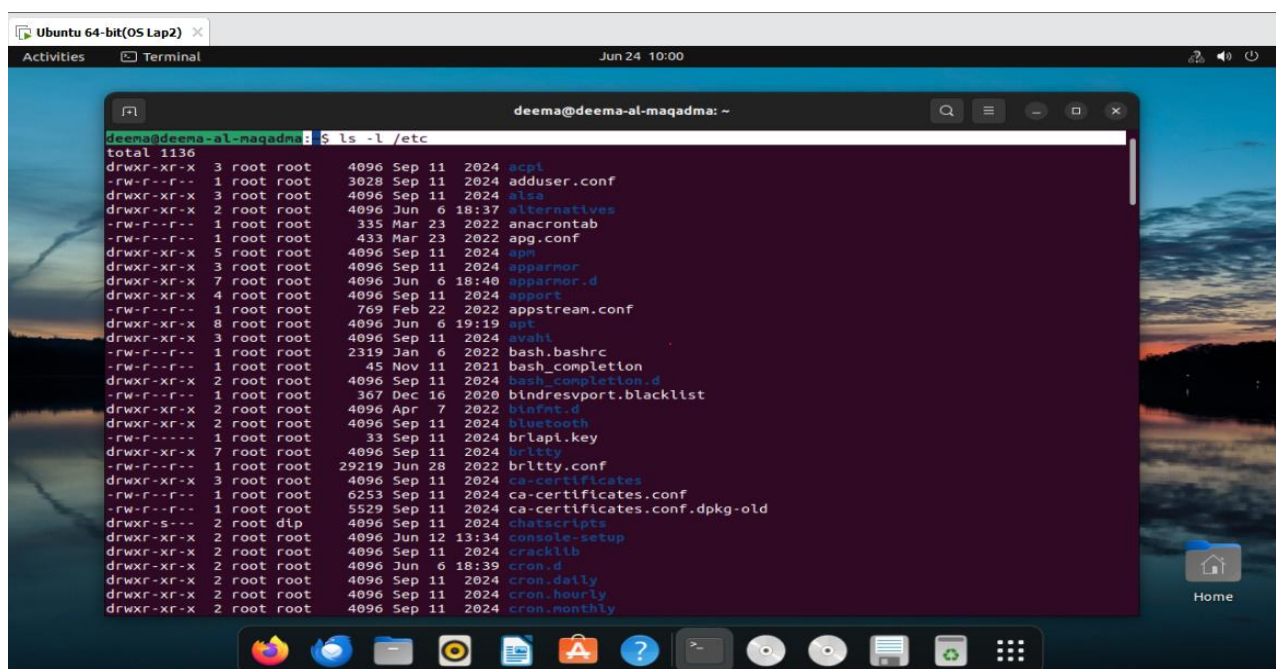


A terminal window titled 'Ubuntu 64-bit(OS Lap2)' showing the command `ls -l /home` and its output. The output lists the /home directory with permissions `drwxr-x---`, owner `deema`, group `deema`, size `4096`, and date `Jun 15 15:22`. The terminal is running on a system named `deema@deema-al-maqadma`.

```
deema@deema-al-maqadma:~$ ls -l /home
total 4
drwxr-x--- 22 deema deema 4096 Jun 15 15:22 deema
deema@deema-al-maqadma:~$
```

List contents of /etc with details

ls -l /etc : Function: Shows system configuration files with their properties, Expected output: Displays files like passwd, hostname, fstab with their details

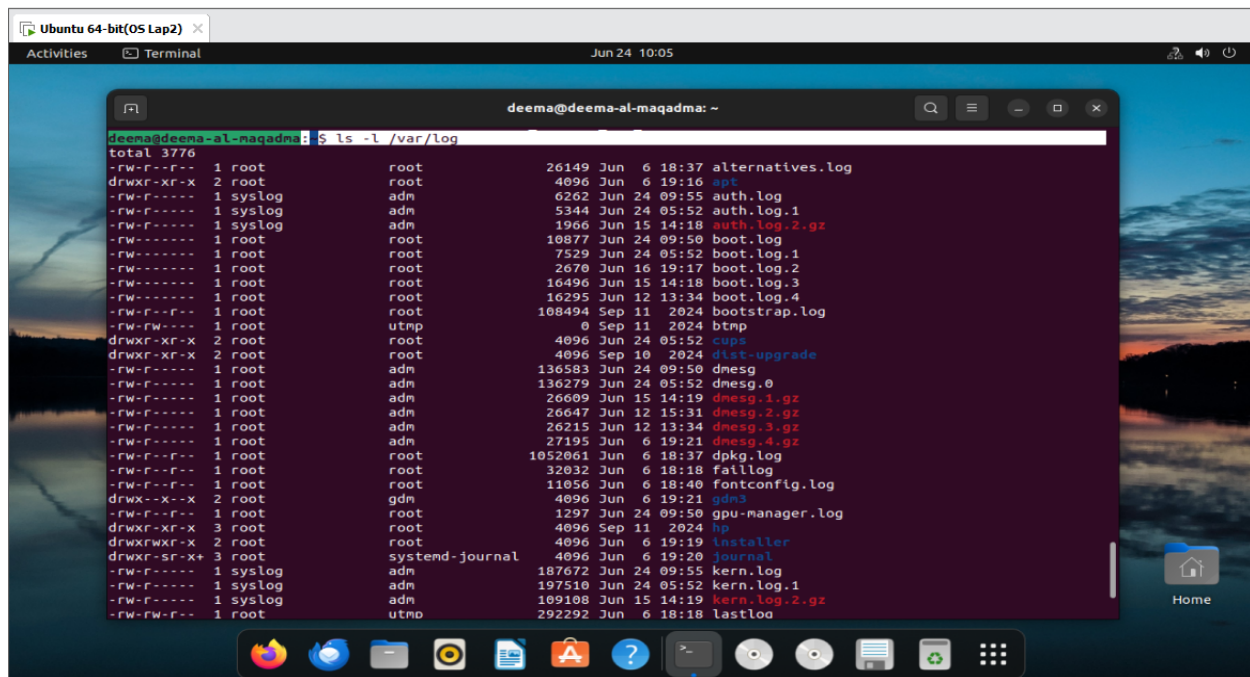


A terminal window titled 'Ubuntu 64-bit(OS Lap2)' showing the command `ls -l /etc` and its output. The output lists various system configuration files in /etc, including `acpi`, `adduser.conf`, `alsa`, `alternatives`, `anacrontab`, `apg.conf`, `apm`, `apparmor`, `apparmor.d`, `appport`, `appstream.conf`, `apt`, `avahi`, `bash.bashrc`, `bash_completion`, `bash_completion.d`, `bindresvport.blacklist`, `binfmt.d`, `bluetooth`, `brlapi.key`, `brltty`, `brltty.conf`, `ca-certificates`, `ca-certificates.conf`, `ca-certificates.conf.dpkg-old`, `chatscripts`, `console-setup`, `cracklib`, `cron.d`, `cron.daily`, `cron.hourly`, and `cron.monthly`. The terminal is running on a system named `deema@deema-al-maqadma`.

```
deema@deema-al-maqadma:~$ ls -l /etc
total 1136
drwxr-xr-x 3 root root 4096 Sep 11 2024 acpi
-rw-r--r-- 1 root root 3028 Sep 11 2024 adduser.conf
drwxr-xr-x 3 root root 4096 Sep 11 2024 alsa
drwxr-xr-x 2 root root 4096 Jun 6 18:37 alternatives
-rw-r--r-- 1 root root 335 Mar 23 2022 anacrontab
-rw-r--r-- 1 root root 433 Mar 23 2022 apg.conf
drwxr-xr-x 5 root root 4096 Sep 11 2024 apm
drwxr-xr-x 3 root root 4096 Sep 11 2024 apparmor
drwxr-xr-x 7 root root 4096 Jun 6 18:40 apparmor.d
drwxr-xr-x 4 root root 4096 Sep 11 2024 appport
-rw-r--r-- 1 root root 769 Feb 22 2022 appstream.conf
drwxr-xr-x 8 root root 4096 Jun 6 19:19 apt
drwxr-xr-x 3 root root 4096 Sep 11 2024 avahi
-rw-r--r-- 1 root root 2319 Jan 6 2022 bash.bashrc
-rw-r--r-- 1 root root 45 Nov 11 2021 bash_completion
drwxr-xr-x 2 root root 4096 Sep 11 2024 bash_completion.d
-rw-r--r-- 1 root root 367 Dec 16 2020 bindresvport.blacklist
drwxr-xr-x 2 root root 4096 Apr 7 2022 binfmt.d
drwxr-xr-x 2 root root 4096 Sep 11 2024 bluetooth
-rw-r--r-- 1 root root 33 Sep 11 2024 brlapi.key
drwxr-xr-x 7 root root 4096 Sep 11 2024 brltty
-rw-r--r-- 1 root root 29219 Jun 28 2022 brltty.conf
drwxr-xr-x 3 root root 4096 Sep 11 2024 ca-certificates
-rw-r--r-- 1 root root 6253 Sep 11 2024 ca-certificates.conf
-rw-r--r-- 1 root root 5529 Sep 11 2024 ca-certificates.conf.dpkg-old
drwxr-xr-x 2 root root 4096 Sep 11 2024 chatscripts
drwxr-xr-x 2 root root 4096 Jun 12 13:34 console-setup
drwxr-xr-x 2 root root 4096 Sep 11 2024 cracklib
drwxr-xr-x 2 root root 4096 Jun 6 18:39 cron.d
drwxr-xr-x 2 root root 4096 Sep 11 2024 cron.daily
drwxr-xr-x 2 root root 4096 Sep 11 2024 cron.hourly
drwxr-xr-x 2 root root 4096 Sep 11 2024 cron.monthly
```

List contents of /var/log

ls -l /var/log : Function: Displays system log files, Expected output: Shows log files such as syslog, auth.log

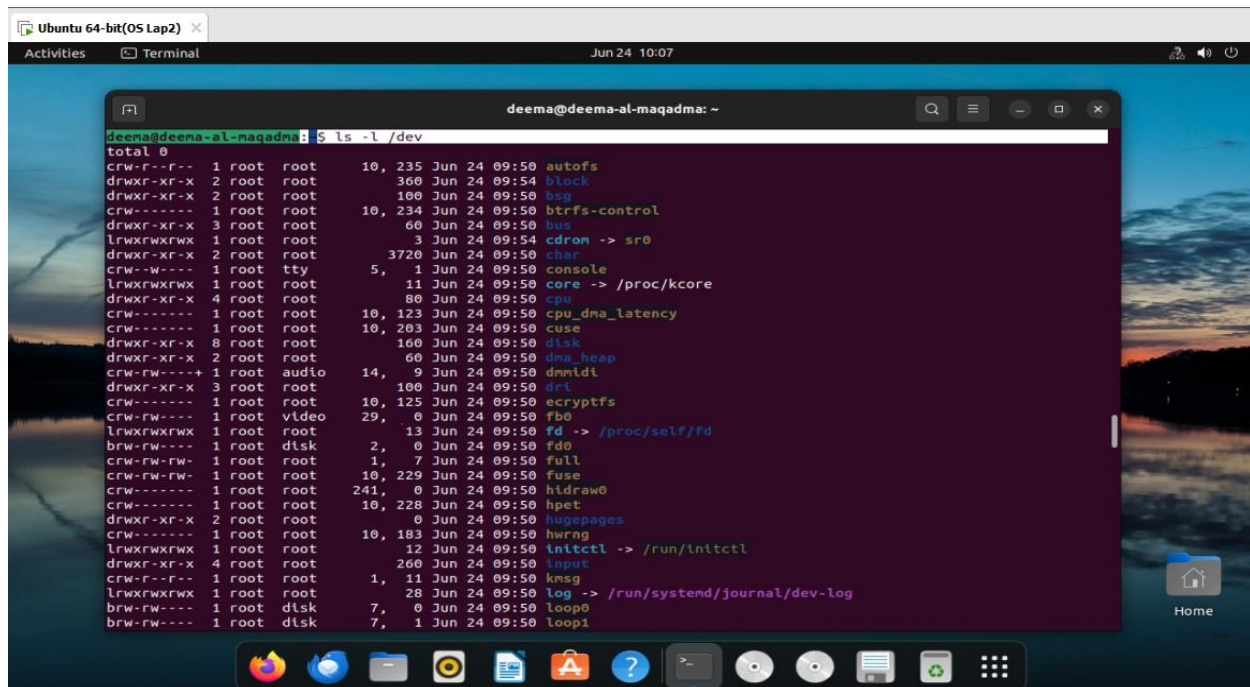


A terminal window titled 'Ubuntu 64-bit(05 Lap2)' showing the command `ls -l /var/log` and its output. The output is a long list of files and directories in /var/log, including alternatives.log, auth.log, boot.log, dmesg, and various log files with their permissions, owners, sizes, and timestamps.

```
deema@deema-al-maqadma: ~  
$ ls -l /var/log  
total 3776  
-rw-r--r-- 1 root root 26149 Jun 6 18:37 alternatives.log  
drwxr-xr-x 2 root root 4096 Jun 6 19:16 aut  
-rw-r----- 1 syslog adm 6262 Jun 24 09:55 auth.log  
-rw-r----- 1 syslog adm 5344 Jun 24 05:52 auth.log.1  
-rw-r----- 1 syslog adm 1966 Jun 15 14:18 auth.log.2.gz  
-rw-r----- 1 root root 10877 Jun 24 09:50 boot.log  
-rw-r----- 1 root root 7529 Jun 24 05:52 boot.log.1  
-rw-r----- 1 root root 2670 Jun 16 19:17 boot.log.2  
-rw-r----- 1 root root 16496 Jun 15 14:18 boot.log.3  
-rw-r----- 1 root root 16295 Jun 12 13:34 boot.log.4  
-rw-r--r-- 1 root root 108494 Sep 11 2024 bootstrap.log  
-rw-rw-r-- 1 root utmp 0 Sep 11 2024 btmp  
drwxr-xr-x 2 root root 4096 Jun 24 05:52 cups  
drwxr-xr-x 2 root root 4096 Sep 10 2024 dist-upgrade  
-rw-r----- 1 root adm 136583 Jun 24 09:50 dmesg  
-rw-r----- 1 root adm 136279 Jun 24 05:52 dmesg.0  
-rw-r----- 1 root adm 26609 Jun 15 14:19 dmesg.1.gz  
-rw-r----- 1 root adm 26647 Jun 12 15:31 dmesg.2.gz  
-rw-r----- 1 root adm 26215 Jun 12 13:34 dmesg.3.gz  
-rw-r----- 1 root adm 27195 Jun 6 19:21 dmesg.4.gz  
-rw-r--r-- 1 root root 1052061 Jun 6 18:37 dpkg.log  
-rw-r--r-- 1 root root 32832 Jun 6 18:18 faillog  
-rw-r--r-- 1 root root 11056 Jun 6 18:40 fontconfig.log  
drwx--x--x 2 root gdm 4096 Jun 6 19:21 gdm3  
-rw-r--r-- 1 root root 1297 Jun 24 09:50 gpu-manager.log  
drwxr-xr-x 3 root root 4096 Sep 11 2024 hp  
drwxrwxr-x 2 root root 4096 Jun 6 19:19 installer  
drwxr-sr-x 3 root systemd-journal 4096 Jun 6 19:20 journal  
-rw-r----- 1 syslog adm 187672 Jun 24 09:55 kern.log  
-rw-r----- 1 syslog adm 197510 Jun 24 05:52 kern.log.1  
-rw-r----- 1 syslog adm 109108 Jun 15 14:19 kern.log.2.gz  
-rw-rw-r-- 1 root utmp 292292 Jun 6 18:18 lastlog
```

List contents of /dev

ls -l /dev : Function: Lists device files connected to the system, Expected output: Displays devices like sda (hard disk), null

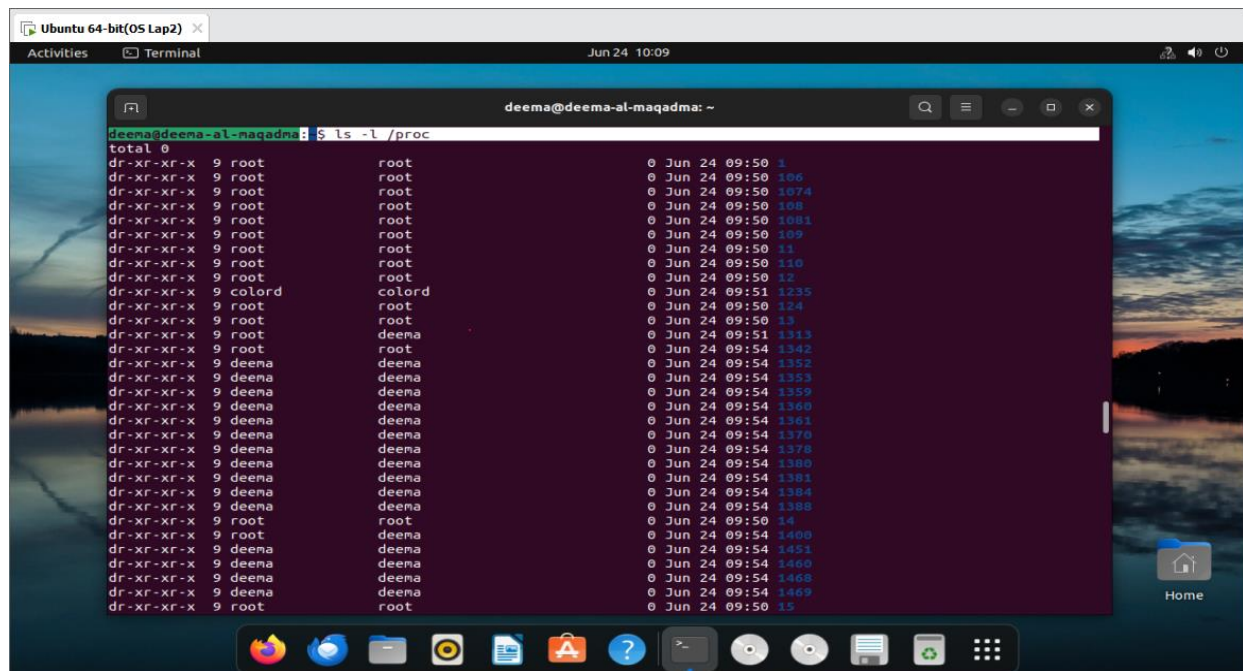


A terminal window titled 'Ubuntu 64-bit(05 Lap2)' showing the command `ls -l /dev` and its output. The output is a long list of device files in /dev, including autofs, block, bsg, btrfs-control, bus, cdrom, char, console, core, cpu, cpu_dma_latency, cuse, disk, dmsetup, dmidekg, dmi, ecryptfs, fb, fd, fd0, full, fuse, hdraw0, hpet, hugetlbfs, hwmon, initctl, input, kmsg, log, loop0, and loop1, with their permissions, owners, sizes, and timestamps.

```
deema@deema-al-maqadma: ~  
$ ls -l /dev  
total 0  
crw-r--r-- 1 root root 10, 235 Jun 24 09:50 autofs  
drwxr-xr-x 2 root root 360 Jun 24 09:54 block  
drwxr-xr-x 2 root root 100 Jun 24 09:50 bsg  
crw-r----- 1 root root 10, 234 Jun 24 09:50 btrfs-control  
drwxr-xr-x 3 root root 60 Jun 24 09:50 bus  
lrwxrwxrwx 1 root root 3 Jun 24 09:54 cdrom -> sr0  
drwxr-xr-x 2 root root 3720 Jun 24 09:50 char  
crw-w--r-- 1 root tty 5, 1 Jun 24 09:50 console  
lrwxrwxrwx 1 root root 11 Jun 24 09:50 core -> /proc/kcore  
drwxr-xr-x 4 root root 80 Jun 24 09:50 cpu  
crw-r----- 1 root root 10, 123 Jun 24 09:50 cpu_dma_latency  
crw-r----- 1 root root 10, 203 Jun 24 09:50 cuse  
drwxr-xr-x 8 root root 160 Jun 24 09:50 disk  
drwxr-xr-x 2 root root 60 Jun 24 09:50 dmsetup  
crw-rw-r-- 1 root audio 14, 9 Jun 24 09:50 dmidekg  
drwxr-xr-x 3 root root 100 Jun 24 09:50 dmi  
crw-r----- 1 root root 10, 125 Jun 24 09:50 ecryptfs  
crw-rw-r-- 1 root video 29, 0 Jun 24 09:50 fb  
lrwxrwxrwx 1 root root 13 Jun 24 09:50 fd -> /proc/self/fd  
brw-rw-r-- 1 root disk 2, 0 Jun 24 09:50 fd0  
crw-rw-rw- 1 root root 1, 7 Jun 24 09:50 full  
crw-rw-rw- 1 root root 10, 229 Jun 24 09:50 fuse  
crw-r----- 1 root root 241, 0 Jun 24 09:50 hdraw0  
crw-r----- 1 root root 10, 228 Jun 24 09:50 hpet  
drwxr-xr-x 2 root root 0 Jun 24 09:50 hugetlbfs  
crw-r----- 1 root root 10, 183 Jun 24 09:50 hwmon  
lrwxrwxrwx 1 root root 12 Jun 24 09:50 initctl -> /run/initctl  
drwxr-xr-x 4 root root 260 Jun 24 09:50 input  
crw-r--r-- 1 root root 1, 11 Jun 24 09:50 kmsg  
lrwxrwxrwx 1 root root 28 Jun 24 09:50 log -> /run/systemd/journal/dev-log  
brw-rw-r-- 1 root disk 7, 0 Jun 24 09:50 loop0  
brw-rw-r-- 1 root disk 7, 1 Jun 24 09:50 loop1
```

List contents of /proc

ls -l /proc : Function: Shows process and system information files, Expected output: Displays virtual files like cpuinfo, meminfo

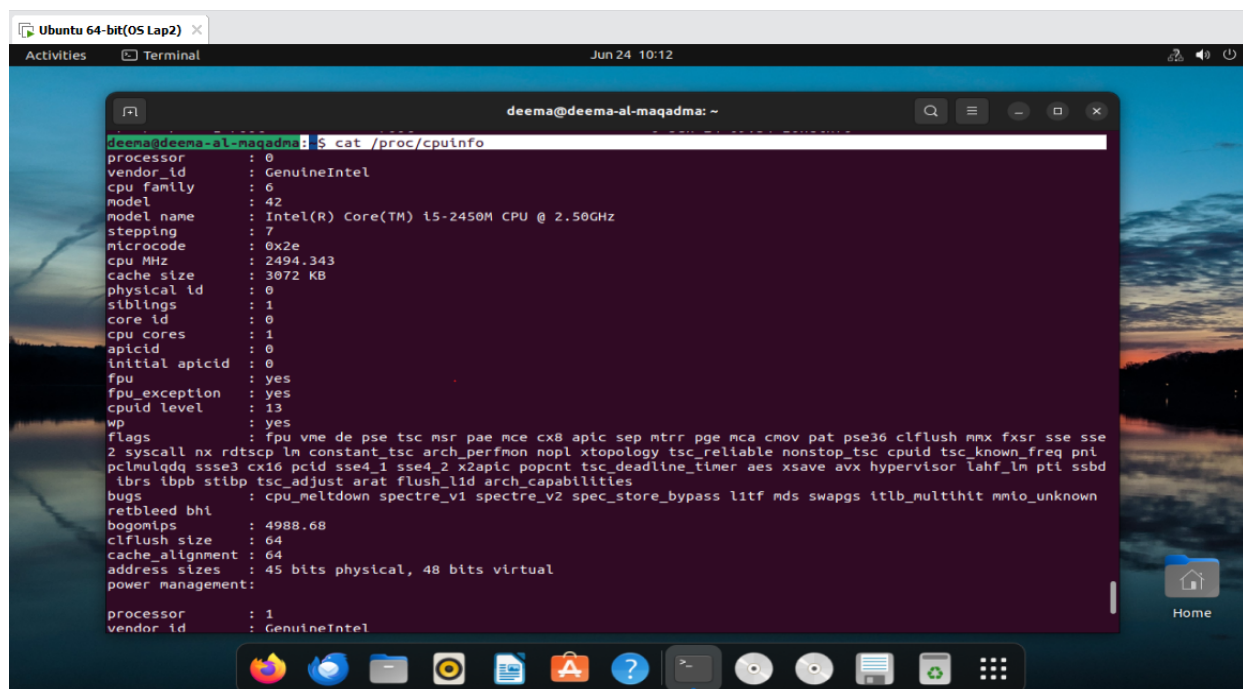


A terminal window titled 'Ubuntu 64-bit(OS Lap2)' showing the command `ls -l /proc` and its output. The output lists various files in /proc, including directories for processes (e.g., /proc/1, /proc/106, /proc/1074) and system information files (e.g., /proc/colord, /proc/deema). The files are listed with permissions, owner, group, size, and timestamps.

```
deema@deema-al-maqadma: ~  
$ ls -l /proc  
total 0  
dr-xr-xr-x 9 root root 0 Jun 24 09:50 1  
dr-xr-xr-x 9 root root 0 Jun 24 09:50 106  
dr-xr-xr-x 9 root root 0 Jun 24 09:50 1074  
dr-xr-xr-x 9 root root 0 Jun 24 09:50 108  
dr-xr-xr-x 9 root root 0 Jun 24 09:50 1081  
dr-xr-xr-x 9 root root 0 Jun 24 09:50 109  
dr-xr-xr-x 9 root root 0 Jun 24 09:50 11  
dr-xr-xr-x 9 root root 0 Jun 24 09:50 110  
dr-xr-xr-x 9 root root 0 Jun 24 09:50 12  
dr-xr-xr-x 9 colord colord 0 Jun 24 09:51 1235  
dr-xr-xr-x 9 root root 0 Jun 24 09:50 124  
dr-xr-xr-x 9 root root 0 Jun 24 09:50 13  
dr-xr-xr-x 9 root deema 0 Jun 24 09:51 1313  
dr-xr-xr-x 9 root root 0 Jun 24 09:54 1342  
dr-xr-xr-x 9 deema deema 0 Jun 24 09:54 1352  
dr-xr-xr-x 9 deema deema 0 Jun 24 09:54 1353  
dr-xr-xr-x 9 deema deema 0 Jun 24 09:54 1359  
dr-xr-xr-x 9 deema deema 0 Jun 24 09:54 1360  
dr-xr-xr-x 9 deema deema 0 Jun 24 09:54 1361  
dr-xr-xr-x 9 deema deema 0 Jun 24 09:54 1370  
dr-xr-xr-x 9 deema deema 0 Jun 24 09:54 1378  
dr-xr-xr-x 9 deema deema 0 Jun 24 09:54 1380  
dr-xr-xr-x 9 deema deema 0 Jun 24 09:54 1381  
dr-xr-xr-x 9 deema deema 0 Jun 24 09:54 1384  
dr-xr-xr-x 9 deema deema 0 Jun 24 09:54 1388  
dr-xr-xr-x 9 root root 0 Jun 24 09:50 14  
dr-xr-xr-x 9 root deema 0 Jun 24 09:54 1400  
dr-xr-xr-x 9 deema deema 0 Jun 24 09:54 1451  
dr-xr-xr-x 9 deema deema 0 Jun 24 09:54 1460  
dr-xr-xr-x 9 deema deema 0 Jun 24 09:54 1468  
dr-xr-xr-x 9 deema deema 0 Jun 24 09:54 1469  
dr-xr-xr-x 9 root root 0 Jun 24 09:50 15
```

Display CPU information

cat /proc/cpuinfo : Function: Shows detailed processor information, Expected output: Displays CPU model, cores, speed



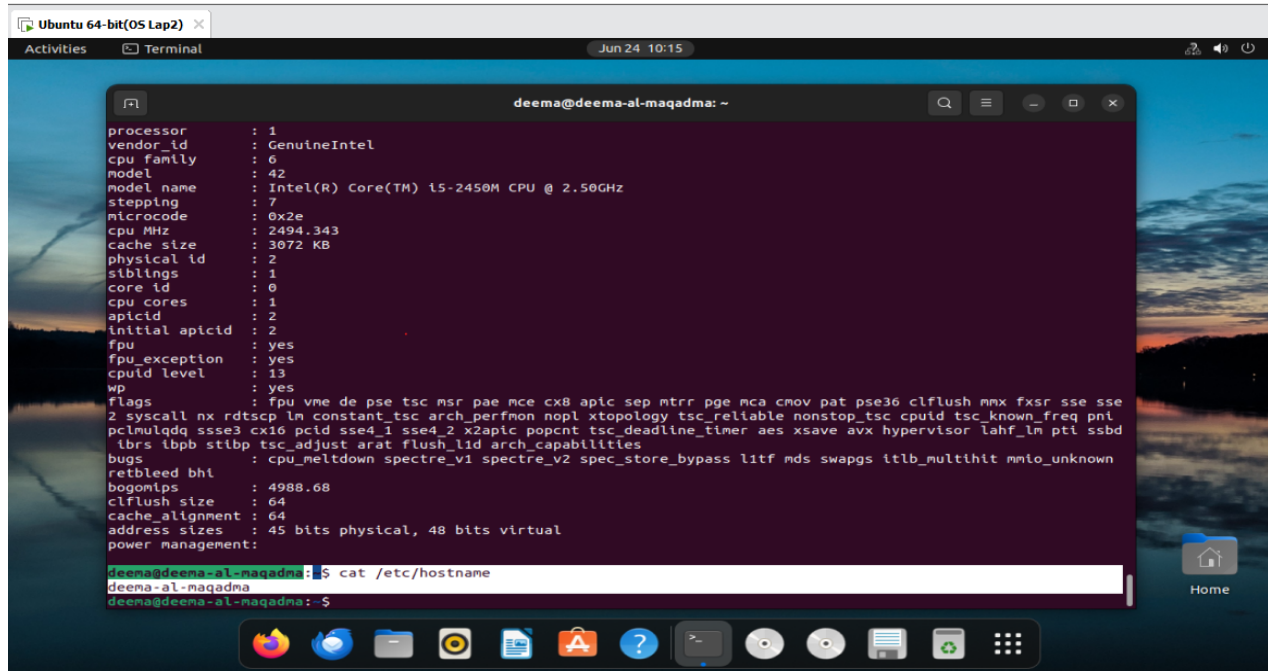
A terminal window titled 'Ubuntu 64-bit(OS Lap2)' showing the command `cat /proc/cpuinfo` and its output. The output displays detailed CPU information for two processors, including vendor, family, model, name, stepping, microcode, frequency, cache size, physical ID, siblings, core ID, CPU cores, apicid, initial apicid, FPU, exception, CPUID level, WP, flags, bugs, bogomips, clflush size, cache alignment, address sizes, and power management.

```
deema@deema-al-maqadma: ~  
$ cat /proc/cpuinfo  
processor       : 0  
vendor_id      : GenuineIntel  
cpu family     : 6  
model          : 42  
model name     : Intel(R) Core(TM) i5-2450M CPU @ 2.50GHz  
stepping       : 7  
microcode      : 0x2e  
cpu MHz        : 2494.343  
cache size     : 3072 KB  
physical id    : 0  
siblings       : 1  
core id        : 0  
cpu cores      : 1  
apicid         : 0  
initial apicid : 0  
fpu            : yes  
fpu_exception  : yes  
cpuid level    : 13  
wp             : yes  
flags           : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2  
                syscall nx rdtscp lm constant_tsc arch_perfmon nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known_freq pni  
                pclmulqdq ssse3 cx16 pcid sse4_1 sse4_2 x2apic popcnt tsc_deadline_timer aes xsave avx hypervisor lahr_lm pti ssbd  
                lbrs lbpb stibp tsc_adjust arat flush_l1d arch_capabilities  
bugs           : cpu_meltdown spectre_v1 spectre_v2 spec_store_bypass l1tf mds swapgs itlb_multihit mmio_unknown  
retbleed bhi  
bogomips       : 4988.68  
clflush size   : 64  
cache alignment : 64  
address sizes  : 45 bits physical, 48 bits virtual  
power management:  
processor       : 1  
vendor_id      : GenuineIntel
```

2. Analyzing Important /etc Files :

View hostname file

cat /etc/hostname : Function: Contains the system's hostname, Expected output: Shows current machine name



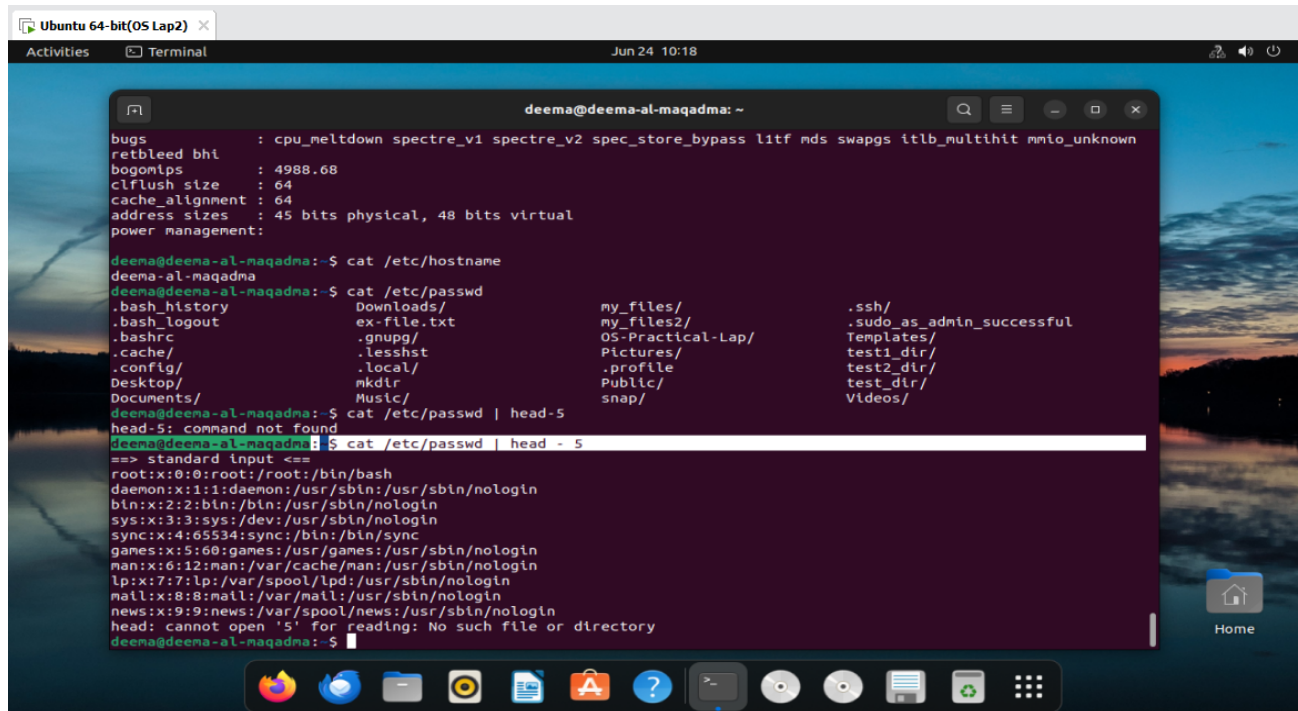
A terminal window titled 'Ubuntu 64-bit(OS Lap2)' showing the output of the `cat /etc/hostname` command. The output displays detailed CPU information, including vendor, model, and various flags. The terminal prompt is `deema@deema-al-maqadma: ~`.

```
processor       : 1
vendor_id      : GenuineIntel
cpu family     : 6
model          : 42
model name     : Intel(R) Core(TM) i5-2450M CPU @ 2.50GHz
stepping       : 7
microcode      : 0x2e
cpu MHz        : 2494.343
cache size     : 3072 KB
physical id    : 2
siblings       : 1
core id        : 0
cpu cores      : 1
apicid         : 2
initial apicid : 2
fpu            : yes
fpu_exception  : yes
cpuid level    : 13
wp             : yes
flags          : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2
                syscall nx rdtscp lm constant_tsc arch_perfmon nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known_freq pni
                pclmulqdq ssse3 cx16 pcid sse4_1 sse4_2 x2apic popcnt tsc_deadline_timer aes xsave avx hypervisor lahf_lm pti ssbd
                ibrs lbrp stibp tsc_adjust arat flush_l1d arch_capabilities
bugs           : cpu_meltdown spectre_v1 spectre_v2 spec_store_bypass l1tf mds swapgs itlb_multihit mmio_unknown
                retbleed bhi
bogomips       : 4988.68
clflush size   : 64
cache_alignment : 64
address sizes   : 45 bits physical, 48 bits virtual
power management:

deema@deema-al-maqadma:~$ cat /etc/hostname
deema-al-maqadma
deema@deema-al-maqadma:~$
```

View first 5 entries in passwd file

cat /etc/passwd | head -5 : Function: Contains user account information, Expected output: Shows first 5 user account records



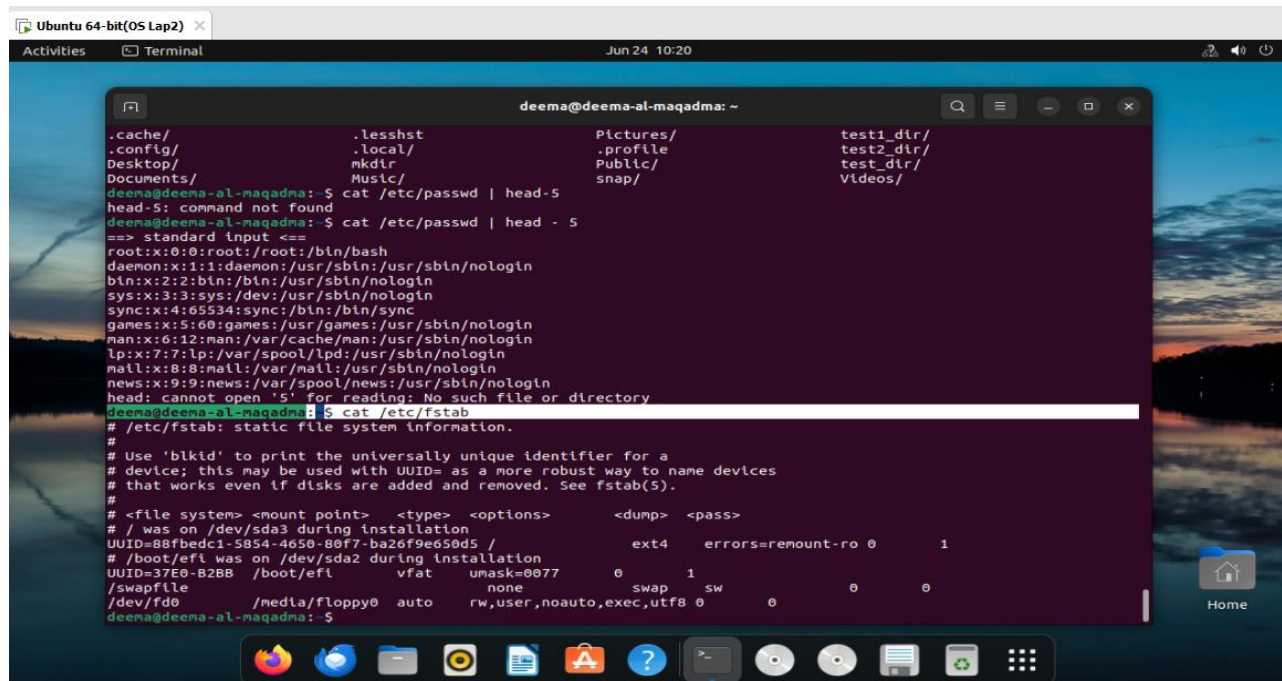
A terminal window titled 'Ubuntu 64-bit(OS Lap2)' showing the output of the `cat /etc/passwd | head -5` command. The output displays the first five user accounts: root, daemon, bin, sys, and sync. The terminal prompt is `deema@deema-al-maqadma: ~`.

```
bugs           : cpu_meltdown spectre_v1 spectre_v2 spec_store_bypass l1tf mds swapgs itlb_multihit mmio_unknown
                retbleed bhi
bogomips       : 4988.68
clflush size   : 64
cache_alignment : 64
address sizes   : 45 bits physical, 48 bits virtual
power management:

deema@deema-al-maqadma:~$ cat /etc/passwd
deema-al-maqadma
deema@deema-al-maqadma:~$ cat /etc/passwd | head -5
head-5: command not found
deema@deema-al-maqadma:~$ cat /etc/passwd | head - 5
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
head: cannot open '5' for reading: No such file or directory
deema@deema-al-maqadma:~$
```

View filesystem table

cat /etc/fstab : Function: Contains filesystem mount information, Expected output: Shows disk partitions and mount points



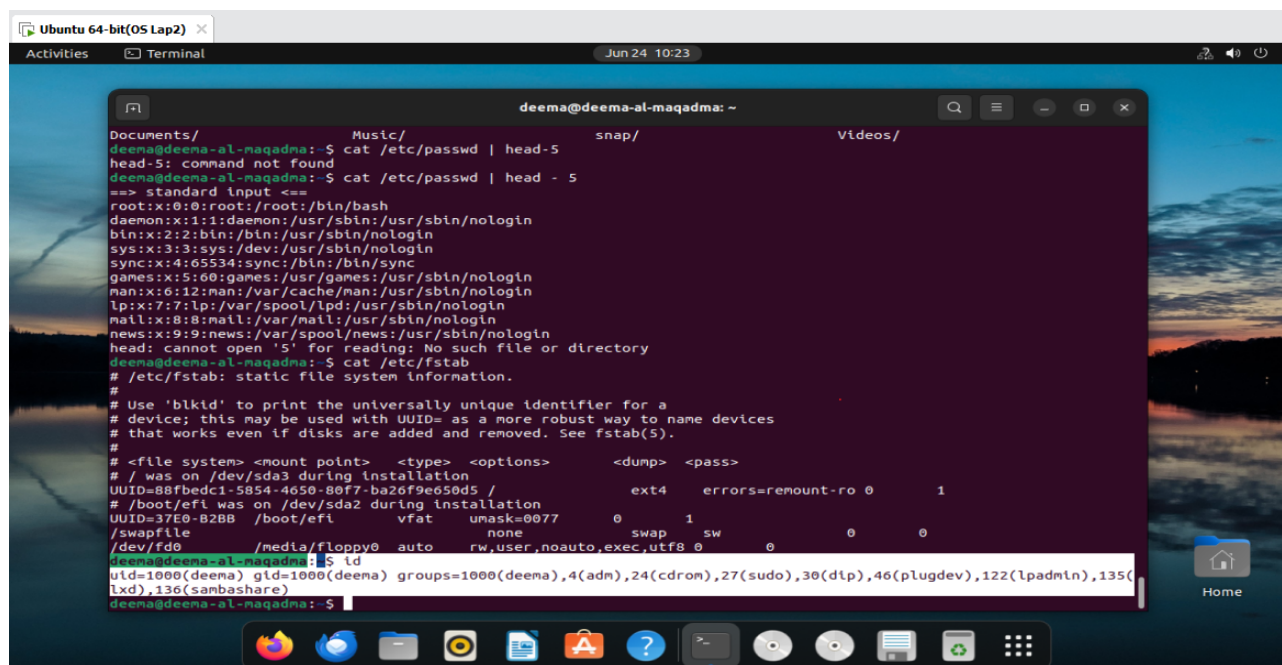
```
deema@deema-al-maqadma: ~  
deema@deema-al-maqadma:~$ cat /etc/passwd | head -5  
head-5: command not found  
deema@deema-al-maqadma:~$ cat /etc/passwd | head - 5  
==> standard input <==  
root:x:0:0:root:/root:/bin/bash  
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin  
bin:x:2:2:bin:/bin:/usr/sbin/nologin  
sys:x:3:3:sys:/dev:/usr/sbin/nologin  
sync:x:4:65534:sync:/bin:/bin/sync  
games:x:5:60:games:/usr/games:/usr/sbin/nologin  
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin  
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin  
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin  
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin  
head: cannot open '5' for reading: No such file or directory  
deema@deema-al-maqadma:~$ cat /etc/fstab  
# /etc/fstab: static file system information.  
#  
# Use 'blkid' to print the universally unique identifier for a  
# device; this may be used with UUID= as a more robust way to name devices  
# that works even if disks are added and removed. See fstab(5).  
#  
# <file system> <mount point> <type> <options> <dump> <pass>  
# / was on /dev/sda3 during installation  
UUID=88fbcd1-5854-4650-80f7-ba26f9e650d5 / ext4 errors=remount-ro 0 1  
# /boot/efi was on /dev/sda2 during installation  
UUID=37E0-B2BB /boot/efi vfat umask=0077 0 1  
/swapfile none swap sw 0 0  
/dev/fd0 /media/floppy0 auto rw,user,noauto,exec,utf8 0 0
```

Part 2: User Analysis \

1. Current User Information :

Show current user information

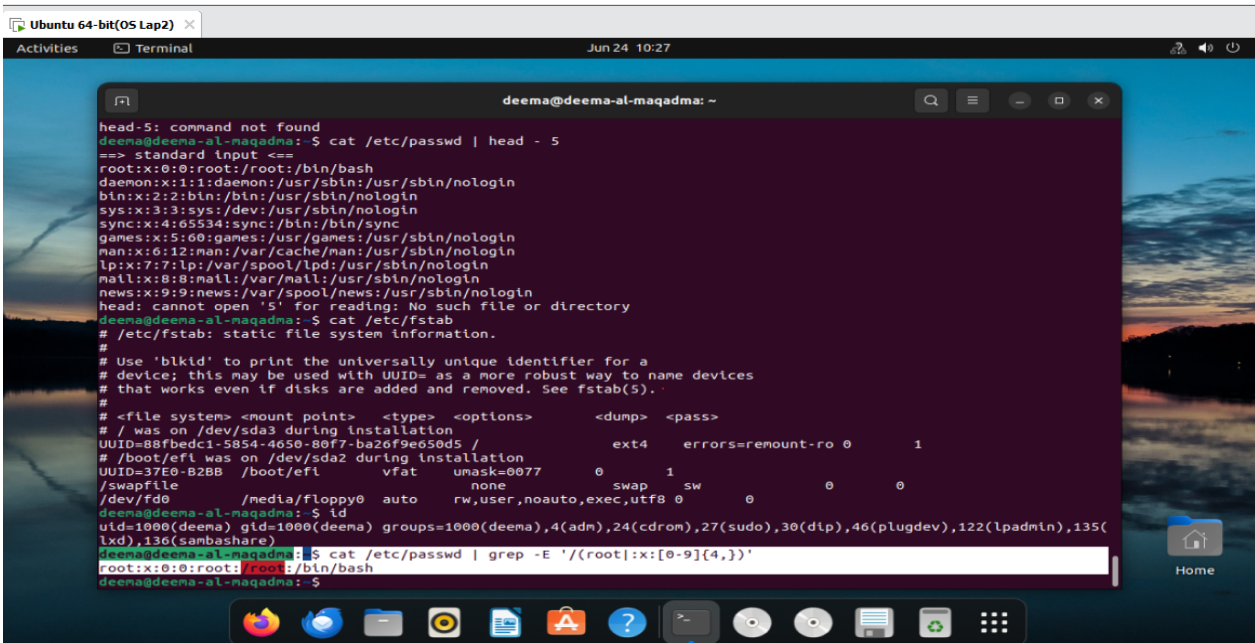
id : - Function: Displays current user's UID, GID and group memberships, Expected output: e.g., uid=1000(student) gid=1000(student) groups=1000(student),4(adm)



```
deema@deema-al-maqadma: ~  
deema@deema-al-maqadma:~$ cat /etc/passwd | head -5  
head-5: command not found  
deema@deema-al-maqadma:~$ cat /etc/passwd | head - 5  
==> standard input <==  
root:x:0:0:root:/root:/bin/bash  
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin  
bin:x:2:2:bin:/bin:/usr/sbin/nologin  
sys:x:3:3:sys:/dev:/usr/sbin/nologin  
sync:x:4:65534:sync:/bin:/bin/sync  
games:x:5:60:games:/usr/games:/usr/sbin/nologin  
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin  
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin  
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin  
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin  
head: cannot open '5' for reading: No such file or directory  
deema@deema-al-maqadma:~$ cat /etc/fstab  
# /etc/fstab: static file system information.  
#  
# Use 'blkid' to print the universally unique identifier for a  
# device; this may be used with UUID= as a more robust way to name devices  
# that works even if disks are added and removed. See fstab(5).  
#  
# <file system> <mount point> <type> <options> <dump> <pass>  
# / was on /dev/sda3 during installation  
UUID=88fbcd1-5854-4650-80f7-ba26f9e650d5 / ext4 errors=remount-ro 0 1  
# /boot/efi was on /dev/sda2 during installation  
UUID=37E0-B2BB /boot/efi vfat umask=0077 0 1  
/swapfile none swap sw 0 0  
/dev/fd0 /media/floppy0 auto rw,user,noauto,exec,utf8 0 0  
deema@deema-al-maqadma:~$ id  
uid=1000(deema) gid=1000(deema) groups=1000(deema),4(adm),24(cdrom),27(sudo),30(dlp),46(plugdev),122(lpadmin),135(lxd),136(sambashare)  
deema@deema-al-maqadma:~$
```

Filter user accounts

cat /etc/passwd | grep -E '/(root|x:[0-9]{4,})' : Function: Shows root and regular user accounts (UID ≥ 1000), Expected output: Displays root and regular user accounts

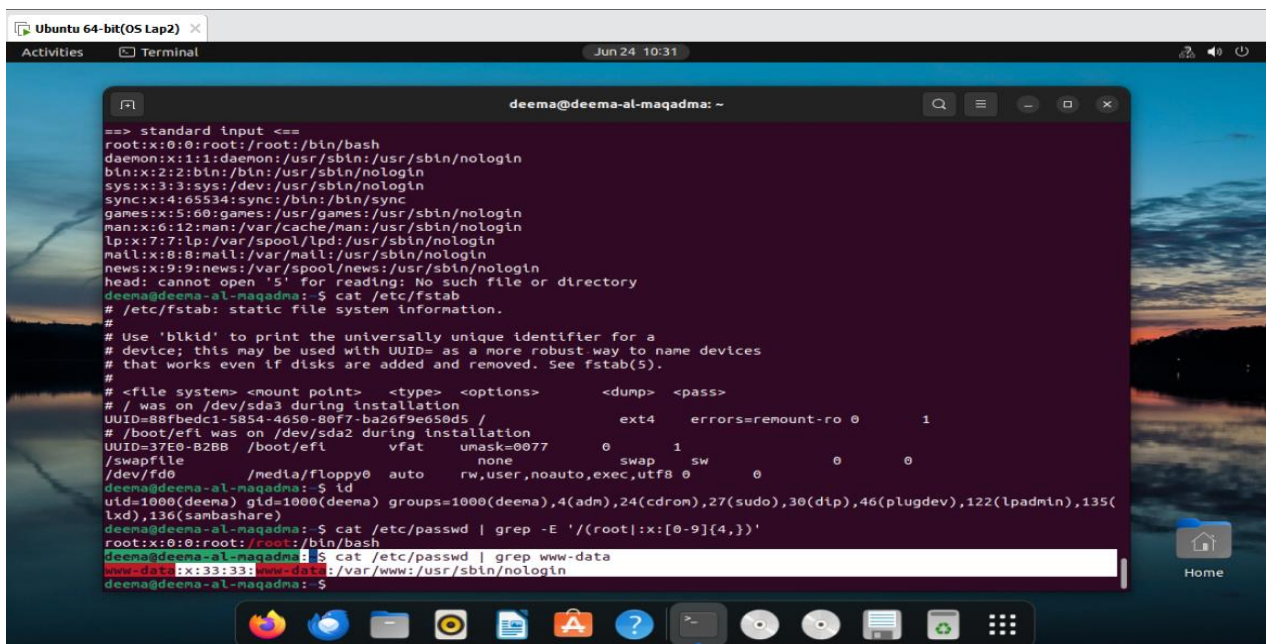


```
deema@deema-al-maqadma: ~  
head-5: command not found  
deema@deema-al-maqadma:~$ cat /etc/passwd | head - 5  
==> standard input <==  
root:x:0:0:root:/root:/bin/bash  
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin  
bin:x:2:2:bin:/bin:/usr/sbin/nologin  
sys:x:3:3:sys:/dev:/usr/sbin/nologin  
sync:x:4:65534:sync:/bin:/bin/sync  
games:x:5:60:games:/usr/games:/usr/sbin/nologin  
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin  
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin  
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin  
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin  
head: cannot open '5' for reading: No such file or directory  
deema@deema-al-maqadma:~$ cat /etc/fstab  
# /etc/fstab: static file system information.  
#  
# Use 'blkid' to print the universally unique identifier for a  
# device; this may be used with UUID= as a more robust way to name devices  
# that works even if disks are added and removed. See fstab(5).  
#  
# <file system> <mount point> <type> <options> <dump> <pass>  
# / was on /dev/sda3 during installation  
UUID=88fbcd1-5854-4050-80f7-ba20f9e050d5 / ext4 errors=remount-ro 0 1  
# /boot/efi was on /dev/sda2 during installation  
UUID=37E0-B2B8 /boot/efi vfat umask=0077 0 1  
/swapfile none swap sw 0 0  
/dev/fd0 /media/floppy0 auto rw,user,noauto,exec,utf8 0 0  
deema@deema-al-maqadma:~$ id  
uid=1000(deema) gid=1000(deema) groups=1000(deema),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),122(lpadmin),135(lxd),136(sambashare)  
deema@deema-al-maqadma:~$ cat /etc/passwd | grep -E '/(root|x:[0-9]{4,})'  
root:x:0:0:root:/root:/bin/bash  
deema@deema-al-maqadma:~$
```

2. System User Analysis :

Find www-data system user

cat /etc/passwd | grep www-data : Function: Shows system user www-data information, Expected output: e.g., www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin



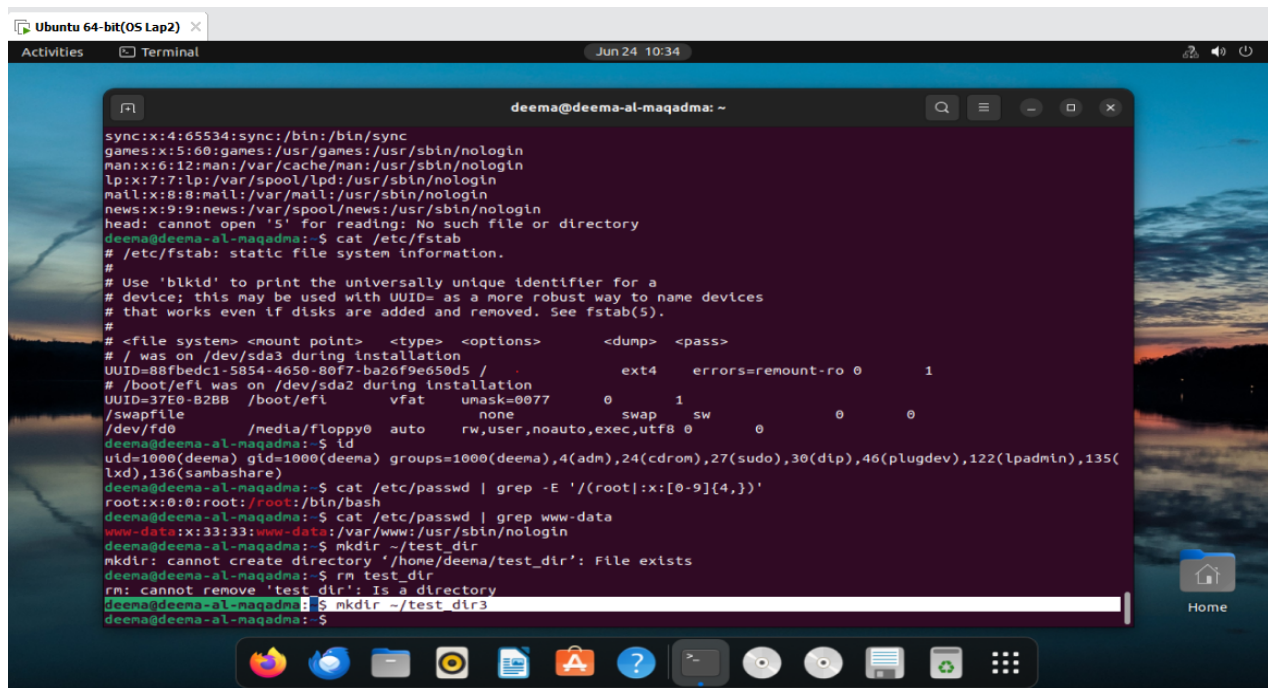
```
deema@deema-al-maqadma: ~  
==> standard input <==  
root:x:0:0:root:/root:/bin/bash  
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin  
bin:x:2:2:bin:/bin:/usr/sbin/nologin  
sys:x:3:3:sys:/dev:/usr/sbin/nologin  
sync:x:4:65534:sync:/bin:/bin/sync  
games:x:5:60:games:/usr/games:/usr/sbin/nologin  
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin  
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin  
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin  
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin  
head: cannot open '5' for reading: No such file or directory  
deema@deema-al-maqadma:~$ cat /etc/fstab  
# /etc/fstab: static file system information.  
#  
# Use 'blkid' to print the universally unique identifier for a  
# device; this may be used with UUID= as a more robust way to name devices  
# that works even if disks are added and removed. See fstab(5).  
#  
# <file system> <mount point> <type> <options> <dump> <pass>  
# / was on /dev/sda3 during installation  
UUID=88fbcd1-5854-4050-80f7-ba20f9e050d5 / ext4 errors=remount-ro 0 1  
# /boot/efi was on /dev/sda2 during installation  
UUID=37E0-B2B8 /boot/efi vfat umask=0077 0 1  
/swapfile none swap sw 0 0  
/dev/fd0 /media/floppy0 auto rw,user,noauto,exec,utf8 0 0  
deema@deema-al-maqadma:~$ id  
uid=1000(deema) gid=1000(deema) groups=1000(deema),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),122(lpadmin),135(lxd),136(sambashare)  
deema@deema-al-maqadma:~$ cat /etc/passwd | grep -E '/(root|x:[0-9]{4,})'  
root:x:0:0:root:/root:/bin/bash  
deema@deema-al-maqadma:~$ cat /etc/passwd | grep www-data  
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin  
deema@deema-al-maqadma:~$
```

Part 3: Creating and Managing Structure \

1. Creating Directory and Files :

Create test_dir in user's home directory

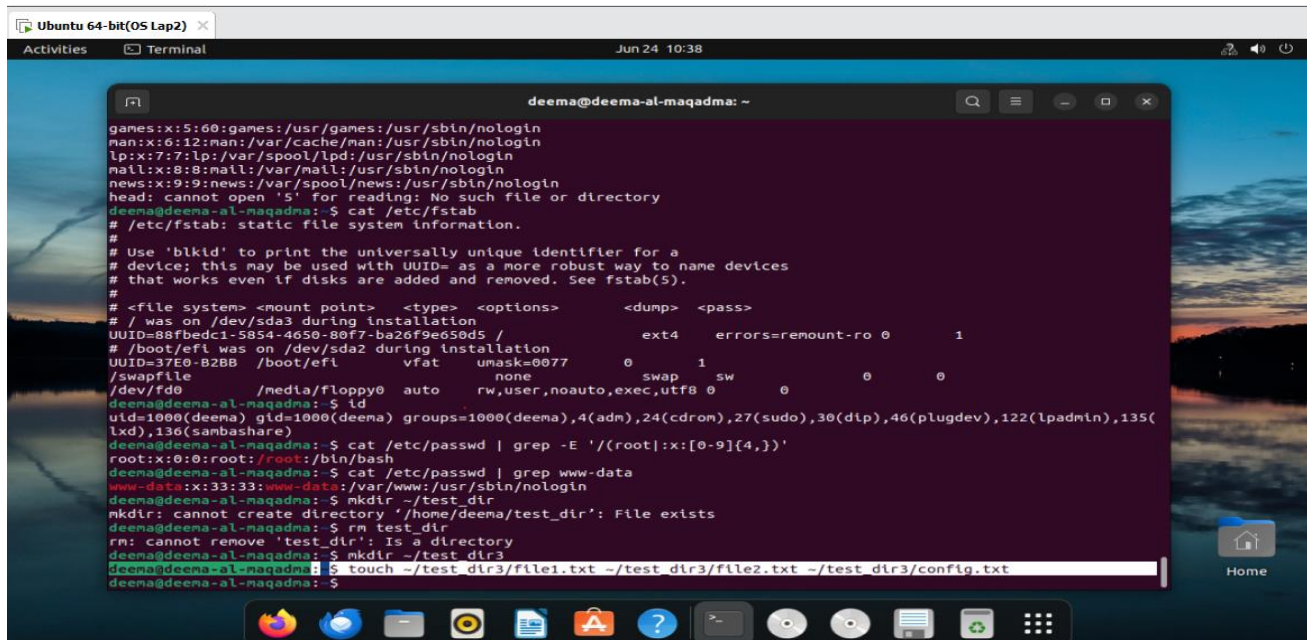
mkdir ~/test_dir : - Function: Creates a new directory named test_dir

A terminal window titled 'Ubuntu 64-bit(OS Lap2)' showing a user named 'deema' at 'deema-al-maqadma'. The terminal output shows the user running 'cat /etc/fstab' and 'id', followed by an attempt to create a directory 'test_dir' in the home directory. The command 'mkdir ~/test_dir' fails with the error 'mkdir: cannot create directory '/home/deema/test_dir': File exists'. The user then runs 'rm test_dir' which fails with 'rm: cannot remove 'test_dir': Is a directory'. Finally, the user runs 'mkdir ~/test_dir3' successfully.

```
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
head: cannot open '5' for reading: No such file or directory
deema@deema-al-maqadma:~$ cat /etc/fstab
# /etc/fstab: static file system information.
#
# Use 'blkid' to print the universally unique identifier for a
# device; this may be used with UUID= as a more robust way to name devices
# that works even if disks are added and removed. See fstab(5).
#
# <file system> <mount point> <type> <options>        <dump> <pass>
# / was on /dev/sda3 during installation
UUID=88fbcd1-5854-4650-80f7-ba26f9e650d5 /      ext4      errors=remount-ro 0      1
# /boot/efi was on /dev/sda2 during installation
UUID=37E0-B2B8 /boot/efi vfat    umask=0077 0      1      0      0
/swapfile /swapfile none    swap      sw      0      0      0
deema@deema-al-maqadma:~$ id
uid=1000(deema) gid=1000(deema) groups=1000(deema),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),122(lpadmin),135(lxd),136(sambashare)
deema@deema-al-maqadma:~$ cat /etc/passwd | grep -E '/(root|:x:[0-9]{4,})'
root:x:0:0:root:/root:/bin/bash
deema@deema-al-maqadma:~$ cat /etc/passwd | grep www-data
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
deema@deema-al-maqadma:~$ mkdir ~/test_dir
mkdir: cannot create directory '/home/deema/test_dir': File exists
deema@deema-al-maqadma:~$ rm test_dir
rm: cannot remove 'test_dir': Is a directory
deema@deema-al-maqadma:~$ mkdir ~/test_dir3
deema@deema-al-maqadma:~$
```

Create 3 empty files in test_dir

touch ~/test_dir/file1.txt ~/test_dir/file2.txt ~/test_dir/config.txt : - Function: Creates three empty text files

A terminal window titled 'Ubuntu 64-bit(OS Lap2)' showing the same user 'deema' at 'deema-al-maqadma'. The terminal output shows the user running 'mkdir ~/test_dir' successfully, followed by 'rm test_dir' which fails with 'rm: cannot remove 'test_dir': Is a directory'. Finally, the user runs 'touch ~/test_dir3/file1.txt ~/test_dir3/file2.txt ~/test_dir3/config.txt' successfully.

```
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
head: cannot open '5' for reading: No such file or directory
deema@deema-al-maqadma:~$ cat /etc/fstab
# /etc/fstab: static file system information.
#
# Use 'blkid' to print the universally unique identifier for a
# device; this may be used with UUID= as a more robust way to name devices
# that works even if disks are added and removed. See fstab(5).
#
# <file system> <mount point> <type> <options>        <dump> <pass>
# / was on /dev/sda3 during installation
UUID=88fbcd1-5854-4650-80f7-ba26f9e650d5 /      ext4      errors=remount-ro 0      1
# /boot/efi was on /dev/sda2 during installation
UUID=37E0-B2B8 /boot/efi vfat    umask=0077 0      1      0      0
/swapfile /swapfile none    swap      sw      0      0      0
deema@deema-al-maqadma:~$ id
uid=1000(deema) gid=1000(deema) groups=1000(deema),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),122(lpadmin),135(lxd),136(sambashare)
deema@deema-al-maqadma:~$ cat /etc/passwd | grep -E '/(root|:x:[0-9]{4,})'
root:x:0:0:root:/root:/bin/bash
deema@deema-al-maqadma:~$ cat /etc/passwd | grep www-data
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
deema@deema-al-maqadma:~$ mkdir ~/test_dir
mkdir: cannot create directory '/home/deema/test_dir': File exists
deema@deema-al-maqadma:~$ rm test_dir
rm: cannot remove 'test_dir': Is a directory
deema@deema-al-maqadma:~$ mkdir ~/test_dir3
deema@deema-al-maqadma:~$ touch ~/test_dir3/file1.txt ~/test_dir3/file2.txt ~/test_dir3/config.txt
deema@deema-al-maqadma:~$
```

2. Setting Permissions :

Set 640 permissions for file1.txt

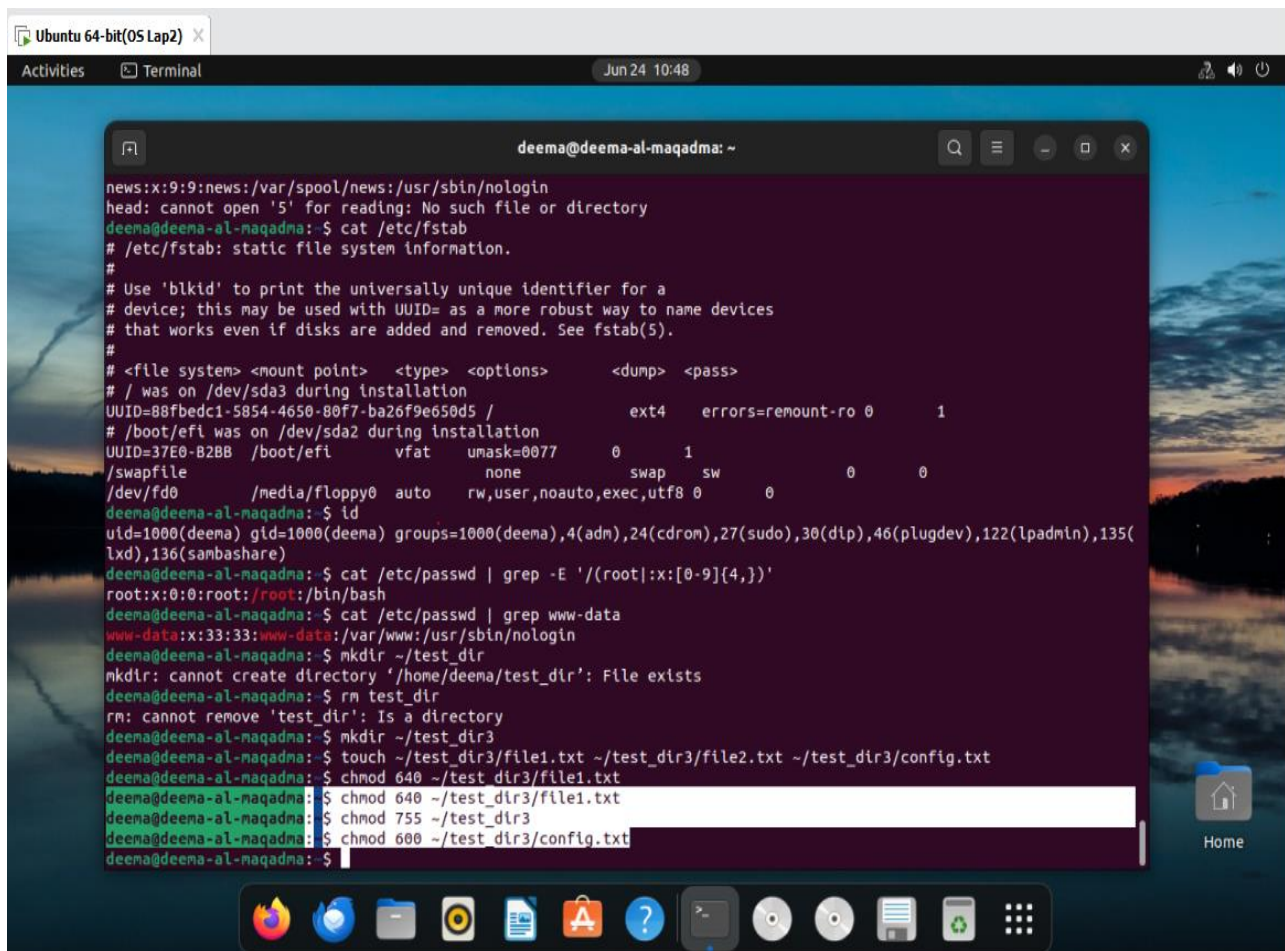
chmod 640 ~/test_dir/file1.txt : Function: Grants owner read+write (6), group read (4), others nothing (0), Explanation: 6=4(read)+2(write), 4=read only, 0=no permissions

Set 755 permissions for test_dir

chmod 755 ~/test_dir: Function: Grants owner all permissions (7), group/others read+execute (5), Explanation: 7=4+2+1(rwx), 5=4+1(r-x)

Set 600 permissions for config.txt

chmod 600 ~/test_dir/config.txt: Function: Grants owner read+write, nothing to group/others, Explanation: 6=4+2(rw-), 0=no group permissions, 0=no others permissions

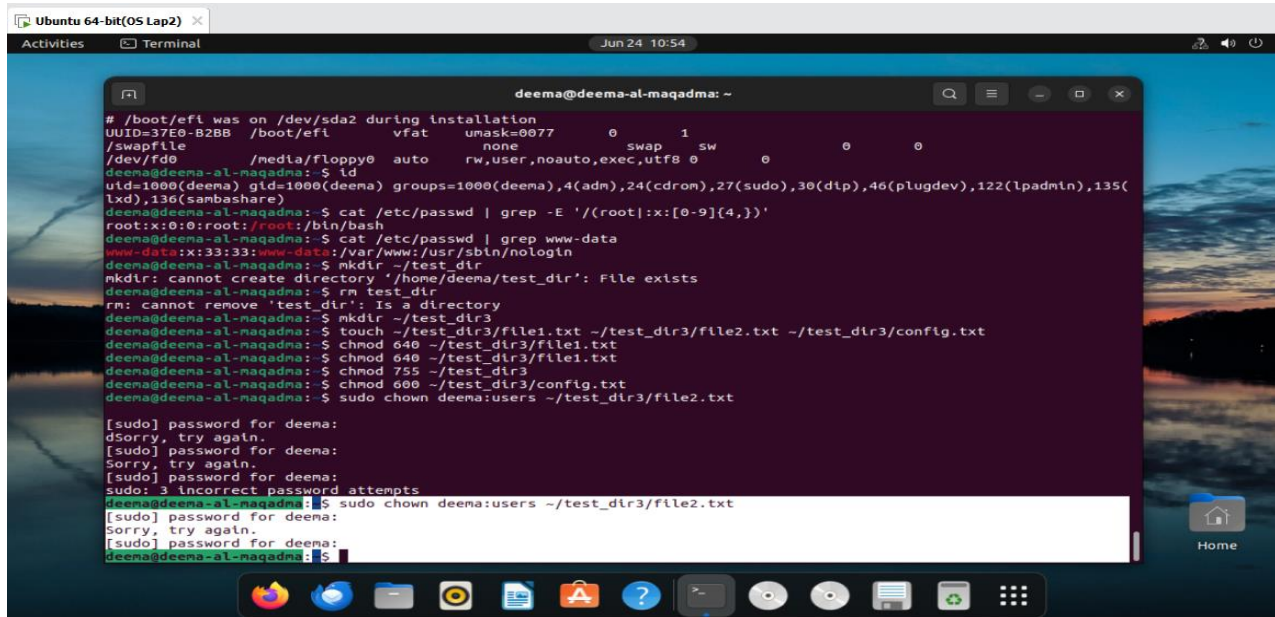


```
deema@deema-al-maqadma: ~  
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin  
head: cannot open '5' for reading: No such file or directory  
deema@deema-al-maqadma:~$ cat /etc/fstab  
# /etc/fstab: static file system information.  
#  
# Use 'blkid' to print the universally unique identifier for a  
# device; this may be used with UUID= as a more robust way to name devices  
# that works even if disks are added and removed. See fstab(5).  
#  
#<file system> <mount point> <type> <options> <dump> <pass>  
# / was on /dev/sda3 during installation  
UUID=88fbdc1-5854-4650-80f7-ba26f9e650d5 / ext4 errors=remount-ro 0 1  
# /boot/efi was on /dev/sda2 during installation  
UUID=37E0-B2B8 /boot/efi vfat umask=0077 0 1  
/swapfile /swapfile none swap sw 0 0  
/dev/fd0 /media/floppy0 auto rw,user,noauto,exec,utf8 0 0  
deema@deema-al-maqadma:~$ id  
uid=1000(deema) gid=1000(deema) groups=1000(deema),4(adn),24(cdrom),27(sudo),30(dip),46(plugdev),122(lpadmin),135(lxd),136(sambashare)  
deema@deema-al-maqadma:~$ cat /etc/passwd | grep -E '/(root|x:[0-9]{4,})'  
root:x:0:0:root:/root:/bin/bash  
deema@deema-al-maqadma:~$ cat /etc/passwd | grep www-data  
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin  
deema@deema-al-maqadma:~$ mkdir ~/test_dir  
mkdir: cannot create directory '/home/deema/test_dir': File exists  
deema@deema-al-maqadma:~$ rm test_dir  
rm: cannot remove 'test_dir': Is a directory  
deema@deema-al-maqadma:~$ mkdir ~/test_dir3  
deema@deema-al-maqadma:~$ touch ~/test_dir3/file1.txt ~/test_dir3/file2.txt ~/test_dir3/config.txt  
deema@deema-al-maqadma:~$ chmod 640 ~/test_dir3/file1.txt  
deema@deema-al-maqadma:~$ chmod 640 ~/test_dir3/file1.txt  
deema@deema-al-maqadma:~$ chmod 755 ~/test_dir3  
deema@deema-al-maqadma:~$ chmod 600 ~/test_dir3/config.txt  
deema@deema-al-maqadma:~$
```

3. Changing Ownership :

Change owner of file2.txt to student:users

sudo chown student:users ~/test_dir/file2.txt: Function: Changes file owner to student and group to users ->Note: May require root privileges (sudo) if not original owner



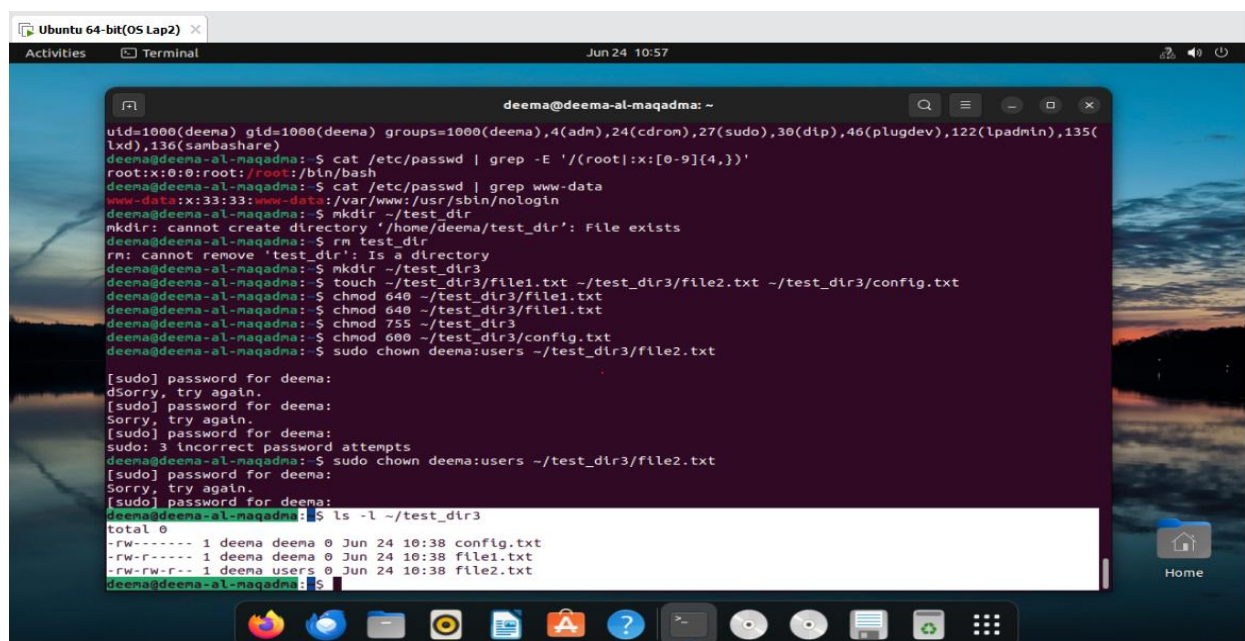
```
deema@deema-al-maqadma: ~  
# /boot/efi was on /dev/sda2 during installation  
UUID=37E0-B2B8 /boot/efi vfat umask=0077 0 swap 1 sw 0 0 0  
/swapfile  
/dev/rfd0 /media/floppy0 auto rw,user,noauto,exec,utf8 0 0 0  
deema@deema-al-maqadma:~$ id  
uid=1000(deema) gid=1000(deema) groups=1000(deema),4(adm),24(cdrom),27(sudo),30(dip),46(plugindev),122(lpadmin),135(lxd),136(sambashare)  
deema@deema-al-maqadma:~$ cat /etc/passwd | grep -E '/(root|x:[0-9]{4,})'  
root:x:0:0:root:/root:/bin/bash  
deema@deema-al-maqadma:~$ cat /etc/passwd | grep www-data  
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin  
deema@deema-al-maqadma:~$ mkdir ~/test_dir  
mkdir: cannot create directory '/home/deema/test_dir': File exists  
deema@deema-al-maqadma:~$ rm test_dir  
rm: cannot remove 'test_dir': Is a directory  
deema@deema-al-maqadma:~$ mkdir ~/test_dir3  
deema@deema-al-maqadma:~$ touch ~/test_dir3/file1.txt ~/test_dir3/file2.txt ~/test_dir3/config.txt  
deema@deema-al-maqadma:~$ chmod 640 ~/test_dir3/file1.txt  
deema@deema-al-maqadma:~$ chmod 755 ~/test_dir3  
deema@deema-al-maqadma:~$ chmod 600 ~/test_dir3/config.txt  
deema@deema-al-maqadma:~$ sudo chown deema:users ~/test_dir3/file2.txt  
[sudo] password for deema:  
dSorry, try again.  
[sudo] password for deema:  
Sorry, try again.  
[sudo] password for deema:  
sudo: 3 incorrect password attempts  
deema@deema-al-maqadma:~$ sudo chown deema:users ~/test_dir3/file2.txt  
[sudo] password for deema:  
Sorry, try again.  
[sudo] password for deema:  
deema@deema-al-maqadma:~$
```

Part 4: Verification and Analysis \

1. Verifying Permissions :

List contents of test_dir with details

ls -l ~/test_dir : Function: Shows files with their permissions and ownership

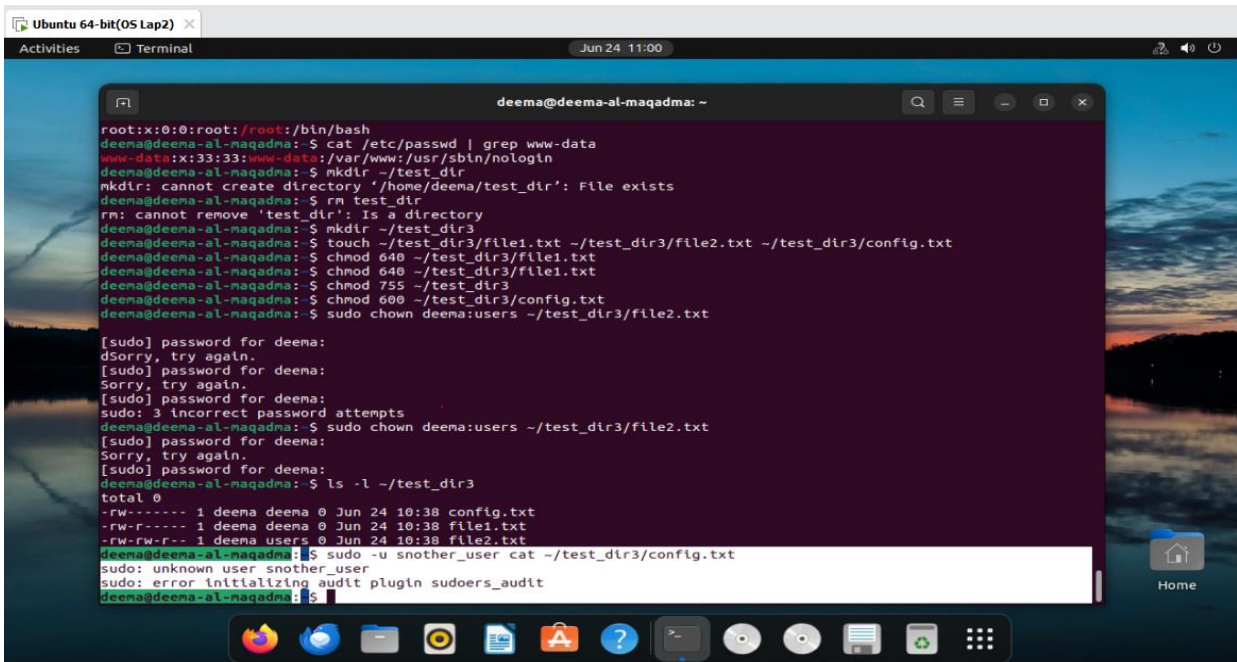


```
deema@deema-al-maqadma: ~  
uid=1000(deema) gid=1000(deema) groups=1000(deema),4(adm),24(cdrom),27(sudo),30(dip),46(plugindev),122(lpadmin),135(lxd),136(sambashare)  
deema@deema-al-maqadma:~$ cat /etc/passwd | grep -E '/(root|x:[0-9]{4,})'  
root:x:0:0:root:/root:/bin/bash  
deema@deema-al-maqadma:~$ cat /etc/passwd | grep www-data  
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin  
deema@deema-al-maqadma:~$ mkdir ~/test_dir  
mkdir: cannot create directory '/home/deema/test_dir': File exists  
deema@deema-al-maqadma:~$ rm test_dir  
rm: cannot remove 'test_dir': Is a directory  
deema@deema-al-maqadma:~$ mkdir ~/test_dir3  
deema@deema-al-maqadma:~$ touch ~/test_dir3/file1.txt ~/test_dir3/file2.txt ~/test_dir3/config.txt  
deema@deema-al-maqadma:~$ chmod 640 ~/test_dir3/file1.txt  
deema@deema-al-maqadma:~$ chmod 755 ~/test_dir3  
deema@deema-al-maqadma:~$ chmod 600 ~/test_dir3/config.txt  
deema@deema-al-maqadma:~$ sudo chown deema:users ~/test_dir3/file2.txt  
[sudo] password for deema:  
dSorry, try again.  
[sudo] password for deema:  
Sorry, try again.  
[sudo] password for deema:  
sudo: 3 incorrect password attempts  
deema@deema-al-maqadma:~$ sudo chown deema:users ~/test_dir3/file2.txt  
[sudo] password for deema:  
Sorry, try again.  
[sudo] password for deema:  
deema@deema-al-maqadma:~$ ls -l ~/test_dir3  
total 0  
-rw-r----- 1 deema deema 0 Jun 24 10:38 config.txt  
-rw-r----- 1 deema deema 0 Jun 24 10:38 file1.txt  
-rw-rw-r-- 1 deema users 0 Jun 24 10:38 file2.txt  
deema@deema-al-maqadma:~$
```

2. Testing config.txt Permissions :

Attempt to read config.txt as another user

sudo -u another_user cat ~/test_dir/config.txt : Function: Tests file access as different user



```
root:x:0:0:root:/root:/bin/bash
deema@deema-al-maqadma:~$ cat /etc/passwd | grep www-data
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
deema@deema-al-maqadma:~$ mkdir ~/test_dir
mkdir: cannot create directory '/home/deema/test_dir': File exists
deema@deema-al-maqadma:~$ rm test_dir
rm: cannot remove 'test_dir': Is a directory
deema@deema-al-maqadma:~$ mkdir ~/test_dir3
deema@deema-al-maqadma:~$ touch ~/test_dir3/file1.txt ~/test_dir3/file2.txt ~/test_dir3/config.txt
deema@deema-al-maqadma:~$ chmod 640 ~/test_dir3/file1.txt
deema@deema-al-maqadma:~$ chmod 640 ~/test_dir3/file2.txt
deema@deema-al-maqadma:~$ chmod 755 ~/test_dir3
deema@deema-al-maqadma:~$ chmod 600 ~/test_dir3/config.txt
deema@deema-al-maqadma:~$ sudo chown deema:users ~/test_dir3/file2.txt
[sudo] password for deema:
dSorry, try again.
[sudo] password for deema:
Sorry, try again.
[sudo] password for deema:
sudo: 3 incorrect password attempts
deema@deema-al-maqadma:~$ sudo chown deema:users ~/test_dir3/file2.txt
[sudo] password for deema:
Sorry, try again.
[sudo] password for deema:
deema@deema-al-maqadma:~$ ls -l ~/test_dir3
total 0
-rw-r----- 1 deema deema 0 Jun 24 10:38 config.txt
-rw-r----- 1 deema deema 0 Jun 24 10:38 file1.txt
-rw-rw-r-- 1 deema users 0 Jun 24 10:38 file2.txt
deema@deema-al-maqadma:~$ sudo -u another_user cat ~/test_dir3/config.txt
sudo: unknown user another_user
sudo: error initializing audit plugin sudoers_audit
deema@deema-al-maqadma:~$
```

Final Report \

Analysis of Permission Settings and User Types :

1. Permission Settings \

- **640** for file1.txt : Gives owner (student) read+write, group (student) read only, and others no access. Suitable for files that should be readable by group members but not modifiable.
- **755** for test_dir : Allows owner full access (rwx), group/others read and execute (enter directory). Standard permissions for shared directories.
- **600** for config.txt: Restricts access to owner only (rw). Essential for sensitive configuration files containing secrets.

2. User Types \

- **Root user (UID 0)**: Has unlimited system privileges for administration tasks.
- **Regular users (UID ≥ 1000)**: Have limited permissions, restricted mainly to their home directories.
- **System users (UID 1-999)**: Service accounts like www-data for web server, with no login capability

"واخر دعواهم ان الحمد لله رب العالمين".